

Paul Parisot

Software Engineer - C++ & Low-Latency Systems

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About

Systems software engineer deeply focused on modern C++ performance optimization, low-level Linux runtimes, and hardware-software boundaries.

Specialized in zero-allocation memory design, concurrent thread synchronization, and asynchronous networking architectures.

Pragmatic debugger experienced in diagnosing subtle performance bottlenecks, memory corruptions, and race conditions in production environments.

Skills

Core Systems: C++ (Modern C++17/20), C, Go, Python, x86/ARM Assembly stubs

Low-Level Linux: Memory Management (**heap**, **stack**, **shm**), POSIX Threads, Atomics, Driver Communication Boundaries

Performance & Networking: Asynchronous I/O (**Boost.Asio**), TCP/IP Paradigms, Cache-Line Linearity (ECS Layout)

Engineering Toolchain: Linux, CMake, GCC, GDB, Valgrind (Memcheck/Helgrind), Git, Jenkins CI/CD pipelines

Professional Experience

Faurecia / Forvia - Software Engineer (Systems & Tooling)

- Developed and optimized low-level C++ software modules tailored for hardware driver communication interfaces.
- Engineered internal testing & visual tracking applications using C# (Unity engine) under strict system resource limitations.
- Analyzed physical architecture constraints to maintain deterministic software behavior and memory guarantees.
- Collaborated closely within an industrial Agile/Scrum framework, tracking requirements and features via **Jira**.

Gno.land - Core Software Engineer (Distributed Runtime Engines)

- Refactored core multi-threaded execution sub-systems to guarantee operational determinism and maximum execution speed.
- Isolated, reproduced, and patched **6 critical bugs** causing runtime crashes, delivering 10k+ LOC to production.
- Authored automated regression and integration testing suites embedded into continuous deployment (CI/CD) workflows.

Epitech / IONIS - Systems Programming Instructor & Code Auditor

- Taught advanced C/C++ memory architectures, pointer arithmetic, and native Linux system programming to **400+ students**.
- Audited large-scale repositories to enforce strict production standards: error-handling paths, cache awareness, and memory safety.

Low-Latency & Systems Technical Projects

rodox – Low-Latency Linux Systems Engine

Built a modular runtime infrastructure in C++ using a data-oriented Entity Component System (ECS) to avoid cache misses. Completely bypassed OS-level heap allocation latencies (**malloc/new**) by designing thread-safe **custom static pool allocators**. Implemented high-throughput, non-blocking asynchronous networking components utilizing **Boost.Asio** over TCP. Hardened execution paths against hardware-software latency spikes using profiling tools including **CMake**, **GDB**, and **Valgrind**.

Skills: Modern C++, SIMD, Boost.Asio, Memory Management, Linux System APIs, Profiling (Valgrind), Concurrency

cclass – Runtime Memory Mutation Engine

Engineered a raw C engine capable of dynamically rewriting compiled assembly instructions inside secure execution memory regions.

Leveraged system-level **mmap** calls with explicit page protections to remove virtual dispatch runtime overheads.

Enforced strict memory alignment checks to prevent hardware execution faults across target host platforms.

Skills: C, Linux Systems, x86 Assembly, mmap, Memory Safety, Low-level Debugging

Sledge – Containerized Orchestration & Core Backend

Architected and implemented a high-performance core backend infrastructure in Go optimized for high-volume execution streams.

Utilized the Docker API at runtime to dynamically spin up, isolate, and orchestrate concurrent environment containers.

Wrote automated validation and monitoring tooling using **Python** scripts to verify continuous data flow and system state health.

Skills: Go, Docker API, Python Scripting, Data Streaming, System Automation

Education

Epitech, Paris

2022 – 2027

Master's Degree in Information Technology (Master's level diploma)

Level-7 Certification - Expert in Software Engineering

Specialization: Systems Architecture

McGill University, Montréal

2025 – 2026

Certificate in Management

Technical Leadership & Strategy